HARSH RATHORE



MERN STACK DEVELOPER | FULL-STACK WEB ENGINEER

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PROFESSIONAL SUMMARY

MERN Stack Architecture | High-Performance Web Systems | AI-Accelerated Delivery

MERN Stack Developer specializing in building scalable web applications with **React (18/19)**, **Node.js**, **Express**, and **MongoDB**. Proven track record engineering low-latency browser applications (<50ms response), modular REST APIs, and commercial e-commerce platforms indexing 10,000+ items. Power-user of modern AI systems (**Antigravity**, **Claude Code**, local LLMs via Ollama) to accelerate feature scaffolding, testing, and full-stack development cycles by 3x. Published researcher with **4 peer-reviewed papers** across international journals (IRJMETS & IJRASET).

EDUCATION

Master of Computer Applications (MCA)
Uka Tarsadia University, Bardoli, Gujarat
Full-Stack Web Architecture, Distributed Systems & Advanced Algorithms — **2026 - 2028**

Bachelor of Computer Applications (BCA)
Uka Tarsadia University, Bardoli, Gujarat
Software Engineering, Database Management & Core CS | **7.772023 - CGPA** — **2026**

TECHNICAL SKILLS

Frontend & Web UI: **React.js (18/19)**, **JavaScript (ES6+)**, HTML5, CSS3, Tailwind CSS, Responsive Web Design, Web Audio API DSP, State Management (Context, Hooks), Electron.

Backend & Databases: **Node.js**, **Express.js**, RESTful API Architecture, PHP, **MongoDB (Mongoose, Document Modeling, Aggregations)**, MySQL (Schema Design & Queries), Session/JWT Auth.

AI Systems & Dev Tools: **Antigravity**, **Claude Code**, Local LLMs (Ollama), Prompt Engineering, AI-Assisted Full-Stack Building, Git & GitHub, Postman, Linux/Bash, Vercel.

Deep Learning & Vision: **PyTorch**, **TensorFlow / Keras**, YOLO11n (Object Detection), CNNs (ResNet-50/18, MobileNetV2, EfficientNet-B0), Fast Fourier Transform (FFT), Model Evaluation & Benchmarking.

FEATURED PROJECTS (PORTFOLIO PLATFORMS)

Dropify — Audiophile Web Streaming & DSP Engine | Flagship Web & Desktop App — GitHub ↗ 2026
Stack: React 18, Web Audio API DSP, Node.js, Express, InnerTube API, Electron, Kotlin (Android), Tailwind CSS

- Architected low-latency web audio streaming platform featuring browser-native **10-band parametric equalizer** (±12dB biquad IIR peaking filters) and mid/side stereo acoustic stage widening.
- Engineered multi-source millisecond-synchronized lyrics resolver (LRCLIB API) and circadian playback auto-mixes; packaged unified cross-platform clients across Web, Desktop Electron, and Android APK with **<50ms stream initiation**.

Shree Karni Textiles — Enterprise E-Commerce Website for Client | Client Production — GitHub ↗ 2026
Stack: PHP, JavaScript (ES6+), MySQL, CSS3, Responsive Design, Composer

- Architected full-stack enterprise e-commerce platform for custom textile distribution, cataloging **10,000+ fabric SKUs** with optimized image delivery pipelines and sub-second page loads.
- Engineered secure session authorization state, transactional cart persistence, and dynamic multi-parameter inventory indexing, handling concurrent wholesale catalog browsing and direct customer order inquiries.

CodeCanvas — Local AI-Assisted Portfolio Builder & UI Workbench | Developer Tooling — GitHub ↗ 2026
Stack: JavaScript (ES6+), HTML5 Canvas, CSS3 Grid/Flexbox Engine, Local AI Code Generation

- Engineered interactive developer creation suite allowing users to visually manipulate and inspect responsive layouts with zero external framework dependencies running at **60 FPS**.
- Integrated local AI code generation (leveraging **Antigravity** and **Claude Code** workflows) to translate natural language design instructions into clean CSS/HTML markup, slashing layout prototyping and site building time by **60%**.

MyVoice — Civic Issue Reporting & Municipal Triage Platform | Civic Tech Platform — GitHub ↗ 2024
Stack: PHP, MySQL, JavaScript (ES6+), RESTful APIs, Session Management, Bootstrap/CSS3

- Engineered civic tech platform enabling citizens to report, geolocate, and track public infrastructure complaints (potholes, garbage/wastage, streetlight failures, drainage) directly to municipal authorities.
- Designed normalized relational MySQL database tracking issue lifecycles (*Reported*, *In Triage*, *Dispatched*, *Resolved*) with role-based admin controls, slashing ticket dispatch overhead by **40%**.

RESEARCH WORKS & PUBLICATIONS (04 PEER-REVIEWED PAPERS)

Deep Learning | Computer Vision | Model Forensics

- **1. Multi-Class Waste Object Detection Using YOLO11n:** Deep learning approach for automated smart municipal waste segregation. Fine-tuned YOLO11n on TACO dataset (1,500 images, 4,784 annotations, 60 categories); achieved 45.63% Precision. Published in **IRJMETS** (Sept 2025/2026) DOI ↗
- **2. Spatial vs. Frequency-Domain CNN Models for Detecting AI-Generated Images:** Comparative study evaluating baseline spatial CNN vs. hybrid Fast Fourier Transform (FFT) across 60,000 images; spatial CNN achieved 85.3% accuracy under real-world compression. Published in **IJRASET** (Dec 2025) DOI ↗
- **3. Effectiveness of Deep Learning Models for Car Model Classification:** Evaluated ResNet50, MobileNetV2, and VGG16 across 7,463 images spanning 16 car models and 8 camera viewpoints; ResNet50 achieved 95.8% accuracy (100% on front-right angles). Published in **IRJMETS** (Sept 2025) DOI ↗
- **4. Performance Benchmarking of CNN Architectures on CIFAR-100:** Controlled evaluation of ResNet-18, EfficientNet-B0, and GoogLeNet trained from scratch without pretraining bias on CIFAR-100 (60,000 images); ResNet-18 achieved 65% top-1 accuracy. Published in **IJRASET** (Nov 2025) DOI ↗